Hence, the equation of normal is $(y-y_1)=\dfrac{(y_1+f)}{(x_1+g)}(x-x_1)$

$$\Rightarrow \quad (y-y_1)(x_1+g)=(y_1+f)(x-x_1)$$

$$\Rightarrow \quad x_1(y-y_1)+g(y-y_1)=y_1(x-x_1)+f(x-x_1)$$

$$\Rightarrow \quad yx_1-xy_1+g(y-y_1)-f(x-x_1)=0\,.$$

Remark

(1) The equation of tangent at (x_1,y_1) to the circle $x^2+y^2=a^2$ is $xx_1+yy_1=a^2$.

(2) The equation of normal at (x_1,y_1) to the circle $x^2+y^2=a^2$ is $xy_1-yx_1=0$.

(3) The normal passes through the centre of the circle.

5.2.3 Condition for the line $y=mx+c$ to be a tangent to the circle $x^2+y^2=a^2$ and finding the point of contact

Let the line $y=mx+c$ touch the circle $x^2+y^2=a^2$. The centre and radius of the circle $x^2+y^2=a^2$ are $(0,0)$ and a respectively.

(i) Condition for a line to be tangent

Then the perpendicular distance of the line $y-mx-c=0$ from $(0,0)$ is

$$\left|\frac{0-m.0-c}{\sqrt{1+m^2}}\right|=\frac{|c|}{\sqrt{1+m^2}}\,.$$

This must be equal to radius .Therefore $\dfrac{|c|}{\sqrt{1+m^2}}=a$ or $c^2=a^2(1+m^2)$.

Thus the condition for the line $y=mx+c$ to be a tangent to the circle $x^2+y^2=a^2$ is $c^2=a^2(1+m^2)$.

(ii) Point of contact

Let (x_1,y_1) be the the point of contact of $y=mx+c$ with the circle $x^2+y^2=a^2$,

Then $y_1=mx_1+c$. ... (1)

Equation of tangent at (x_1,y_1) is $xx_1+yy_1=a^2$.

$$yy_1=-xx_1+a^2 \qquad \text{... (2)}$$

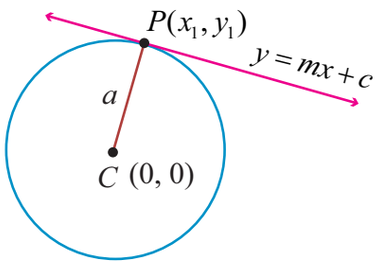


Fig.5.14

Equations (1) and (2) represent the same line and hence the coefficients are proportional.

$$\text{So,}\quad \frac{y_1}{1}=\frac{-x_1}{m}=\frac{a^2}{c}$$

$$y_1=\frac{a^2}{c}, x_1=\frac{-a^2m}{c},\ c=\pm a\sqrt{1+m^2}\,.$$

Then the points of contact is either $\left(\dfrac{-am}{\sqrt{1+m^2}},\dfrac{a}{\sqrt{1+m^2}}\right)$

$$\text{or}\ \left(\frac{am}{\sqrt{1+m^2}},\frac{-a}{\sqrt{1+m^2}}\right).$$

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